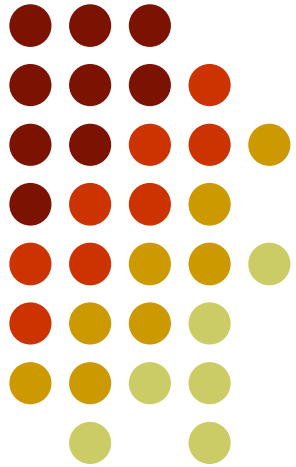
6 现代投资组合理论



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概要

- 均值-方差偏好
- 均值-方差前沿组合
- 均值-方差前沿组合的性质
- 存在无风险资产的情形

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6.1均值-方差偏好

- 如果个体对未来财富分布的偏好可以用财富分布的前两阶矩（均值和方差）来刻画

$$E\left[u\left(\tilde{w}\right)\right]=v\left(E\left[\tilde{w}\right],\sigma^{2}\left(\tilde{w}\right)\right)$$

则称投资者具有**均值-方差偏好**

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6.1均值-方差偏好

- 均值-方差偏好是否具有一般性？
- 例子
  - 假定个体初始财富为1
  - 假定经济中存在两个组合P_1和P_2，随机回报率分别为

$$\tilde{r}_1 = \begin{cases} 1/2, p = 1/2 \\ -1/2, p = 1/2 \end{cases}, \quad \tilde{r}_2 = \begin{cases} 1/3, p = 2/3 \\ -2/3, p = 1/3 \end{cases}$$

  - 均值-方差偏好下的选择？
  - 效用函数取$u(w) = \ln w$的选择？

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6.1均值-方差偏好

- 令 $\bar{w} \equiv E[\tilde{w}],\ \sigma_w^2 \equiv E\left[(\tilde{w}-\bar{w})^2\right]$

$$u(\tilde{w}) = u(\bar{w}) + u'(\bar{w})(\tilde{w}-\bar{w}) + \frac{1}{2}u''(\bar{w})(\tilde{w}-\bar{w})^2 + \cdots \frac{1}{n!}u^{(n)}(\bar{w})(\tilde{w}-\bar{w})^n$$

$$E[u(\tilde{w})] = u(\bar{w}) + \frac{1}{2}u''(\bar{w})\sigma_w^2 + E[R_3]$$

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6.1均值–方差偏好

- 定理6.1 如果$u(\bullet)$是二次的，那么参与者的偏好具有均值-方差表述

$$E\left[u\left(\tilde{w}\right)\right]=v\left(\overline{w},\sigma_{w}\right)$$

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定理6.1证明

- 如果$u(\cdot)$是二次的，不妨将其写成

$$u(\tilde{w}) = \tilde{w} - \frac{1}{2}b\tilde{w}^2, b > 0$$

$$E\left[u(\tilde{w})\right] = E\left[\tilde{w}\right] - \frac{1}{2}bE\left[\tilde{w}^2\right]$$

$$= E\left[\tilde{w}\right] - \frac{1}{2}b\left(\left(E\left[\tilde{w}\right]\right)^2 + Var\left[\tilde{w}\right]\right)$$

$$= \left(1 - \frac{1}{2}b\bar{w}\right)\bar{w} - \frac{1}{2}b\left(\sigma_w\right)^2$$

$$= v\left(\bar{w}, \sigma_w\right)$$

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6.1均值–方差偏好

- 定理6.2 如果证券的支付是联合正态分布的，那么参与者的偏好具有均值-方差表述

$$E\left[u\left(\tilde{w}\right)\right]=v\left(\overline{w},\sigma_{w}\right)$$

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定理6. 2证明

如果证券的支付 $\tilde{X}$ 是由联合正态分布的随机变量组成的向量

则对于任意组合 θ， $\tilde{w} = \tilde{X}\theta$ 也是正态分布的

定义 $\tilde{e} = \dfrac{\tilde{w} - \overline{w}}{\sigma_w}$

那么 $\tilde{e} \sim_d N(0,1)$ 是一个标准正态随机变量且 $\tilde{w} = \overline{w} + \sigma_w \tilde{e}$

立即有 $E\left[u(\tilde{w})\right] = v(\overline{w}, \sigma_w)$

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6.1均值-方差偏好

- 定理6.3 如果证券支付是联合正态分布的，并且

$u'(\bullet)>0$ 以及 $u''(\bullet)<0$，那么均值-方差效用函数具有如下性质

（1） $\partial_1 v>0$；（2） $\partial_2 v<0$；

（3） $\partial_{11}^2 v<0$；（4） $\partial_{22}^2 v<0$；

（5） $v(\overline{w},\sigma_w)$ 是凹的。

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定理6. 3证明

(1) 因为 $u'(\bullet)>0$，$\partial_1 v = E\left[u'\left(\overline{w}+\sigma_w\tilde{e}\right)\right]>0$

(2) $\partial_2 v = E\left[u'\left(\overline{w}+\sigma_w\tilde{e}\right)\tilde{e}\right]$，因为 $u''(\bullet)<0$

对于 $\tilde{e}>0, \overline{w}+\sigma_w\tilde{e}>\overline{w} \Rightarrow u'\left(\overline{w}+\sigma_w\tilde{e}\right)<u'\left(\overline{w}\right)$，有 $u'\left(\overline{w}+\sigma_w\tilde{e}\right)\tilde{e}<u'\left(\overline{w}\right)\tilde{e}$

对于 $\tilde{e}<0, \overline{w}+\sigma_w\tilde{e}<\overline{w} \Rightarrow u'\left(\overline{w}+\sigma_w\tilde{e}\right)>u'\left(\overline{w}\right)$，有 $u'\left(\overline{w}+\sigma_w\tilde{e}\right)\tilde{e}<u'\left(\overline{w}\right)\tilde{e}$

因此 $E\left[u'\left(\overline{w}+\sigma_w\tilde{e}\right)\tilde{e}\right]<E\left[u'\left(\overline{w}\right)\tilde{e}\right]=0$

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定理6. 3证明

(3) 因为 $u^{"}(\bullet)<0$，$\partial_{11}^2 v = E\left[u^{"}\left(\bar{w}+\sigma_w \tilde{e}\right)\right]<0$

(4) 因为 $u^{"}(\bullet)<0$，$\partial_{22}^2 v = E\left[u^{"}\left(\bar{w}+\sigma_w \tilde{e}\right)\tilde{e}^2\right]<0$

(5) v函数的Hessian矩阵顺序主子式

$$H_1 = \partial_{11}^2 v < 0$$

$$H_2 = \partial_{11}^2 v \times \partial_{22}^2 v - \left(\partial_{12}^2 v\right)^2$$

$$= E\left[u^{"}\left(\tilde{w}\right)\right]E\left[u^{"}\left(\tilde{w}\right)\tilde{e}^2\right] - \left(E\left[u^{"}\left(\tilde{w}\right)\tilde{e}\right]\right)^2 \geq 0$$

Hessian矩阵负定，故v是凹的。

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6.2均值-方差前沿组合

- 证券收益向量

$$\tilde{r} = \left[\tilde{r}_1; \tilde{r}_2; \cdots; \tilde{r}_N\right]$$

- 均值向量

$$\bar{r} = E\left[\tilde{r}\right] = \left[E\left(\tilde{r}\right)_1; E\left(\tilde{r}_2\right); \cdots; E\left(\tilde{r}_N\right)\right]$$

- 方差-协方差矩阵

$$V = E\left[\left(\tilde{r} - \bar{r}\right)\left(\tilde{r} - \bar{r}\right)^T\right]$$

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6. 2均值–方差前沿组合

- 组合权重向量 $z=\left[z_1;z_2;\cdots;z_N\right]$
- 组合收益率 $\tilde{r}_Z=z^T\tilde{r}$
- 组合均值 $\overline{r}_Z=E\left[\tilde{r}_Z\right]=E\left[z^T\tilde{r}\right]=z^T\overline{r}$
- 组合方差 $\sigma_Z^2=E\left[\tilde{r}_Z-\overline{r}_Z\right]^2=E\left[z^T\left(\tilde{r}-\overline{r}\right)\right]^2=z^TVz$

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6. 2均值–方差前沿组合

$$\min \frac{1}{2} z^T V z$$

$$s.t.\ \ z^T \bar{r} = \bar{r}_p$$

$$z^T \vec{1} = 1$$

- 上述问题的解称为均值-方差前沿组合(mean-variance frontier portfolio)

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6. 2均值-方差前沿组合

- 假设N只证券线性独立，则V是正定矩阵

$$L = \frac{1}{2} z^T V z + \lambda_r (r_p - z^T \bar{r}) + \lambda_1 (1 - z^T \vec{1})$$

- 一阶条件为

$$\begin{cases} Vz - \lambda_r \bar{r} - \lambda_1 \vec{1} = 0 & (1) \\ \bar{r}_p - z^T \bar{r} = 0 & (2) \\ 1 - z^T \vec{1} = 0 & (3) \end{cases}$$

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6. 2均值–方差前沿组合

$$z = V^{-1}\left(\lambda_r \bar{r} + \lambda_1 \vec{1}\right) \quad (4)$$

$$\bar{r}_p = \left(\lambda_r \bar{r}^T V^{-1} + \lambda_1 \vec{1}^T V^{-1}\right)\bar{r} = \lambda_r \bar{r}^T V^{-1}\bar{r} + \lambda_1 \vec{1}^T V^{-1}\bar{r} \quad (5)$$

$$1 = \left(\lambda_r \bar{r}^T V^{-1} + \lambda_1 \vec{1}^T V^{-1}\right)\vec{1} = \lambda_r \bar{r}^T V^{-1}\vec{1} + \lambda_1 \vec{1}^T V^{-1}\vec{1} \quad (6)$$

令 $a = \vec{1}^T V^{-1}\bar{r} = \bar{r}^T V^{-1}\vec{1}$, $b = \bar{r}^T V^{-1}\bar{r}$, $c = \vec{1}^T V^{-1}\vec{1}$, $d = bc - a^2$

$$\lambda_r = \frac{c\bar{r}_p - a}{d}$$

$$\lambda_1 = \frac{b - a\bar{r}_p}{d}$$

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6. 2均值–方差前沿组合

$$z = \lambda_r V^{-1}\bar{r} + \lambda_1 V^{-1}\vec{1}$$

$$= \frac{1}{d}\left[bV^{-1}\vec{1} - aV^{-1}\bar{r}\right] + \frac{1}{d}\left[cV^{-1}\bar{r} - aV^{-1}\vec{1}\right]\bar{r}_p$$

- 令 $\vec{g} = \frac{1}{d}\left(bV^{-1}\vec{1} - aV^{-1}\bar{r}\right)$, $\vec{h} = \frac{1}{d}\left(cV^{-1}\bar{r} - aV^{-1}\vec{1}\right)$

$$z = \vec{g} + \vec{h}\bar{r}_p$$

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6. 2均值–方差前沿组合



- 定理6.4 任何一个具有均值-方差偏好的参与者的最优组合是一个均值-方差前沿组合。

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6. 3均值-方差前沿组合的性质



- 定理6.5 任意MVF组合可以由z_0和z_1两个MVF组合组合而成
- 证明：

取 $\overline{r}_p = 0$，则 $z_0 = \vec{g}$

取 $\overline{r}_p = 1$，则 $z_1 = \vec{g} + \vec{h}$

对 $\forall \overline{r}_p$，$z_p = \vec{g} + \vec{h}\overline{r}_p = z_0 + (z_1 - z_0)\overline{r}_p = (1 - \overline{r}_p) z_0 + \overline{r}_p z_1$

为 z_0 和 z_1 的线性组合，权重为 $(1 - \overline{r}_p, \overline{r}_p)$

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6. 3均值–方差前沿组合的性质



- 推论6.1 均值-方差前沿组合可由任意两个不同MVF组合而成
- 推论6.2 MVF的任意组合也是MVF组合

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6. 3均值-方差前沿组合的性质

- 例子

$$\bar{r} = \begin{bmatrix} \bar{r}_1 \\ \bar{r}_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}; \quad V = \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}$$

- 求解均值-方差前沿组合

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6. 3均值–方差前沿组合的性质



- 定理6.6 均值-方差偏好下两基金分离成立

- 定理6.7 均值-方差前沿组合位于 R^N 中的一条直线上

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6. 3均值-方差前沿组合的性质

- 任意两个MVF组合p和q的协方差

$$\begin{aligned}
&\operatorname{cov}\left(\tilde{r}_p, \tilde{r}_q\right) = z_p^T V z_q \\
&= z_p^T V\left[V^{-1}\left(\lambda_q \bar{r} + \lambda_1 \vec{1}\right)\right] \\
&= \lambda_q \bar{r}_p + \lambda_1 \\
&= \frac{c\bar{r}_q - a}{d}\bar{r}_p + \frac{b - a\bar{r}_q}{d} \\
&= \frac{1}{d}\left[c\bar{r}_p\bar{r}_q - a\left(\bar{r}_p + \bar{r}_q\right) + b\right] \\
&= \frac{c}{d}\left(\bar{r}_p - \frac{a}{c}\right)\left(\bar{r}_q - \frac{a}{c}\right) + \frac{1}{c}
\end{aligned}$$

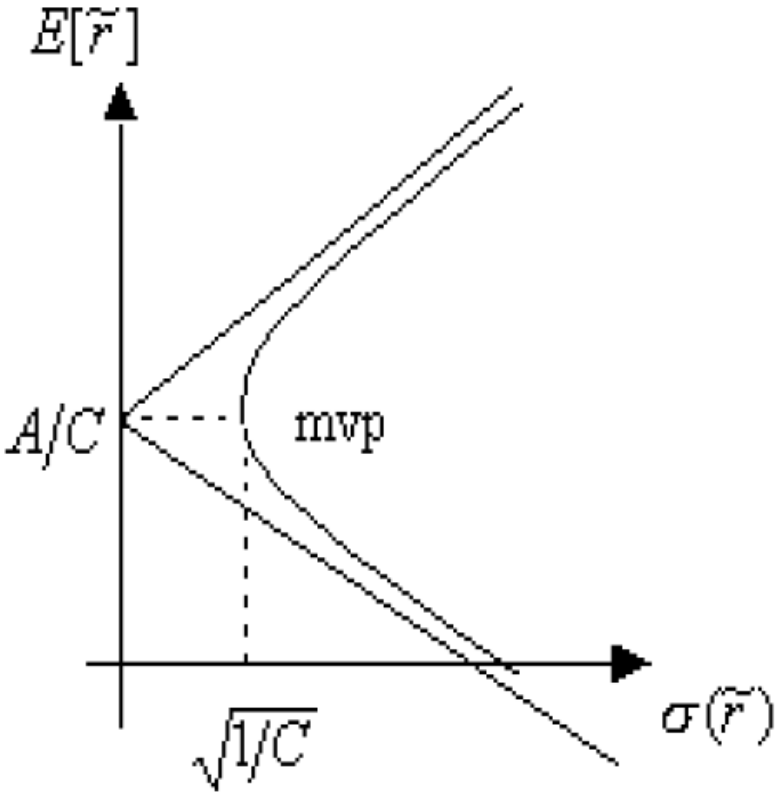
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6. 3均值-方差前沿组合的性质

- 取q=p，则有

$$\sigma_p^2 = \frac{c}{d}\left[\bar{r}_p - \frac{a}{c}\right]^2 + \frac{1}{c}$$

- 全局最小方差组合MVP



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6. 3均值–方差前沿组合的性质

- 定理6.8 对于任意组合p，$\mathrm{cov}\left(\tilde{r}_p, \tilde{r}_{MVP}\right) = \sigma^2_{MVP}$

- 定理6.9 对于任意MVF组合p（$p \neq MVP$），存在一个MVF组合zcp使得

$$\mathrm{cov}\left(\tilde{r}_p, \tilde{r}_{zcp}\right) = 0$$

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6. 3均值–方差前沿组合的性质



- 定理6.10 令p为任一MVF组合。对于任意组合q，我们有

$$\bar{r}_q - \bar{r}_{zcp} = \beta_{q,p}\left(\bar{r}_p - \bar{r}_{zcp}\right)$$

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定理6.10证明

- 由p的最优性，有一阶条件 $z_p^T V = \lambda_r \bar{r}^T + \lambda_1 \vec{1}^T$

$$cov\left(\tilde{r}_p, \tilde{r}_q\right) = z_p^T V z_q = \lambda_r \bar{r}^T z_q + \lambda_1 \vec{1}^T z_q = \lambda_r \bar{r}_q + \lambda_1$$

取 q=p，有 $\bar{r}_p = -\frac{\lambda_1}{\lambda_r} + \frac{\sigma_p^2}{\lambda_r}$　　取 q=zcp，有 $\bar{r}_{zcp} = -\frac{\lambda_1}{\lambda_r}$

$$\bar{r}_p - \bar{r}_{zcp} = \frac{\sigma_p^2}{\lambda_r}$$

$$\bar{r}_q = \bar{r}_{zcp} + \frac{\bar{r}_p - \bar{r}_{zcp}}{\sigma_p^2} cov\left(\tilde{r}_q, \tilde{r}_p\right) = \bar{r}_{zcp} + \beta_{q,p}\left(\bar{r}_p - \bar{r}_{zcp}\right)$$

6. 3均值–方差前沿组合的性质

- 例子
  - 3个风险资产，未来4个等概率的可能状态

	$\tilde{r}_1$	$\tilde{r}_2$	$\tilde{r}_3$
ω_1	2	1	0
ω_2	0	1	2
ω_3	0	2	0
ω_4	2	2	2

  - 求解均值-方差前沿组合

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6. 4存在无风险资产的情形

- 无风险证券F，无风险利率r_F
- 风险证券的超额收益向量 $\bar{\eta} = \bar{r} - r_F \vec{1}$

- 优化问题

$$\min \frac{1}{2} z^T V z$$

$$s.t.\ \ z^T \left(\bar{r} - r_F \vec{1} \right) = z^T \bar{\eta} = \bar{\eta}_p$$

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6.4存在无风险资产的情形

- 拉格朗日函数 $L=\frac{1}{2}z^TVz+\lambda\left(\bar{\eta}_p-z^T\bar{\eta}\right)$
- 一阶条件 $\begin{cases}Vz-\lambda\bar{\eta}=0\\ \bar{\eta}_p-z^T\bar{\eta}=0\end{cases}$

$$z_p=\lambda V^{-1}\bar{\eta}=\frac{\bar{\eta}_p}{\bar{\eta}^TV^{-1}\bar{\eta}}V^{-1}\bar{\eta}$$

对于 $\bar{\eta}_p=0$， $z_p=0$ 无风险资产

对于 $\bar{\eta}_p=\frac{\bar{\eta}^TV^{-1}\bar{\eta}}{\vec{1}^TV^{-1}\bar{\eta}}$， $\vec{1}^Tz_p=1$ 切点组合（tangent portfolio），记作 z_T

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6. 4存在无风险资产的情形

- 对于任意MVF组合z_p，$\vec{1}^T z_p = a_p \neq 1$

$$a_p = \vec{1}^T z_p = \vec{1}^T \frac{\bar{\eta}_p}{\bar{\eta}^T V^{-1} \bar{\eta}} V^{-1} \bar{\eta} = \frac{\bar{\eta}_p}{\frac{\bar{\eta}^T V^{-1} \bar{\eta}}{\vec{1}^T V^{-1} \bar{\eta}}} = \frac{\bar{\eta}_p}{\bar{\eta}^T z_T} = \frac{\bar{\eta}_p}{\bar{\eta}_T}$$

即 $z_p = a_p z_T$

任意 MVF 组合是无风险证券和切点组合 z_T 的线性组合

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6. 4存在无风险资产的情形



- **定理6.11** 当存在无风险证券时，MVF可由无风险证券和切点组合组合而成

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6. 4存在无风险资产的情形

$$\sigma_p^2 = z_p^T V z_p$$

$$= \left(\lambda V^{-1}\bar{\eta}\right)^T V \left(\lambda V^{-1}\bar{\eta}\right)$$

$$= \lambda^2 \bar{\eta}^T V^{-1}\bar{\eta}$$

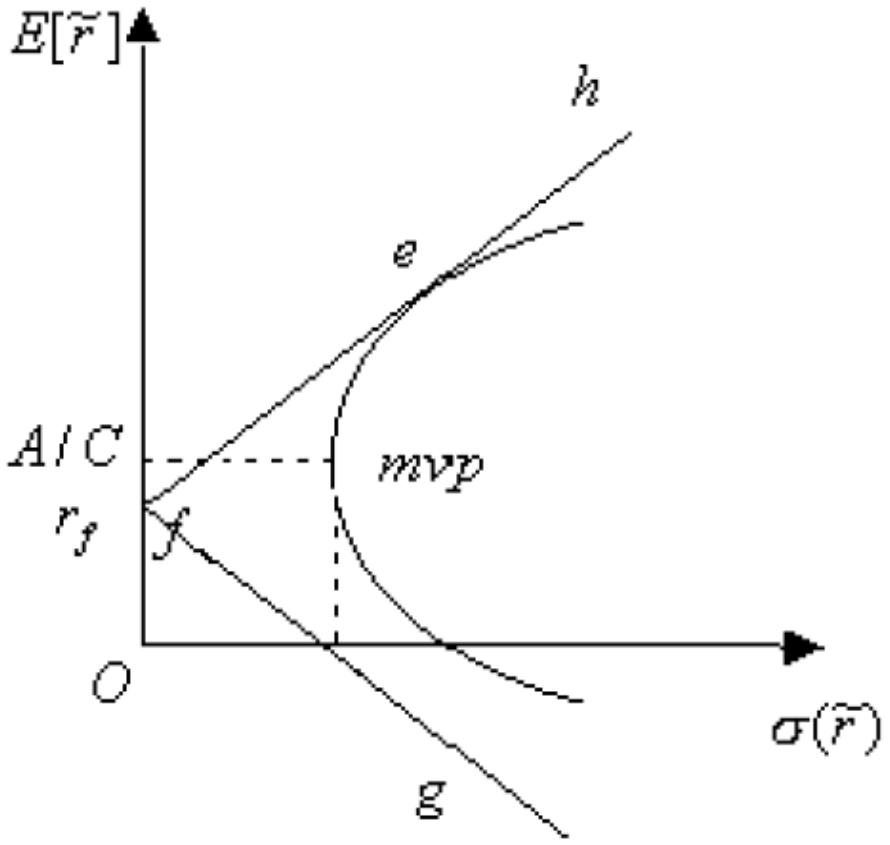
$$= \left(\frac{\bar{\eta}_p}{\bar{\eta}^T V^{-1}\bar{\eta}}\right)^2 \bar{\eta}^T V^{-1}\bar{\eta}$$

$$= \frac{\bar{\eta}_p^2}{\bar{\eta}^T V^{-1}\bar{\eta}}$$

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6. 4存在无风险资产的情形

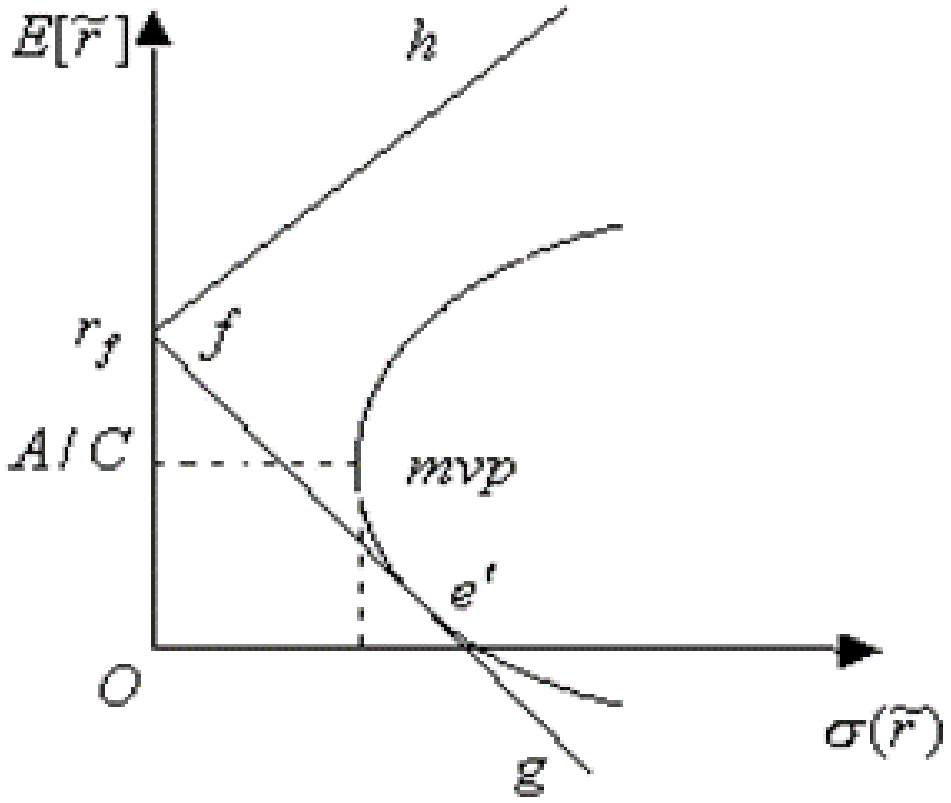
$$r_F < \frac{a}{c}$$



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6. 4存在无风险资产的情形

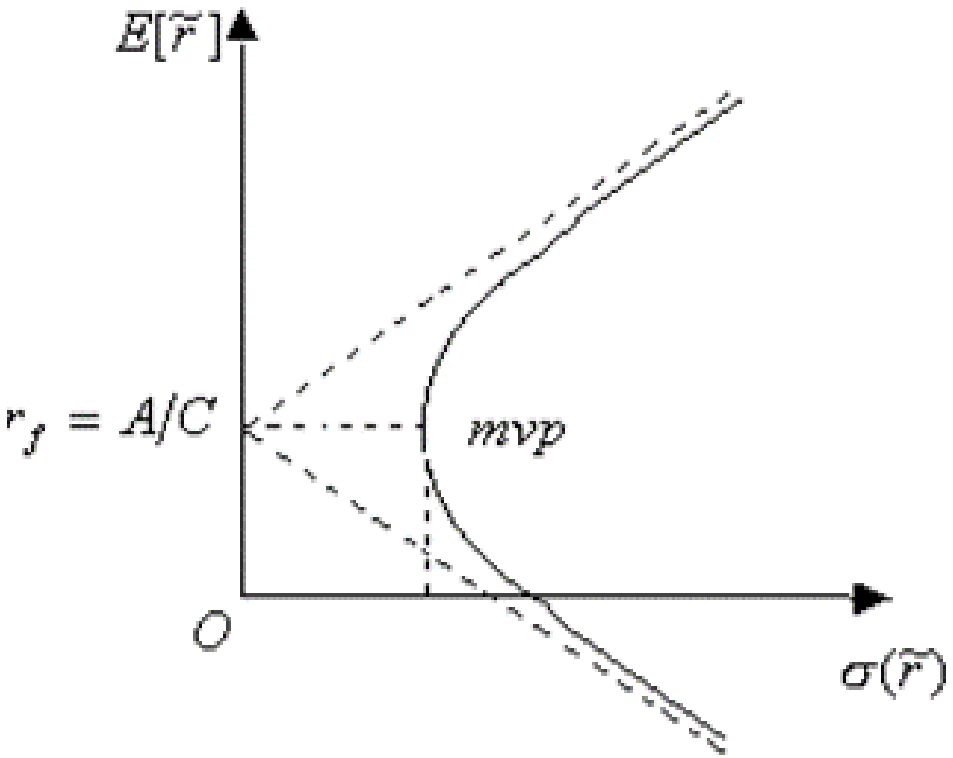
$$r_F > \frac{a}{c}$$



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6. 4存在无风险资产的情形

$$r_F = \frac{a}{c}$$



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6. 4存在无风险资产的情形

- 定理6.12 当 $r_F < \overline{r}_{MVP}$ 时，切点组合是一个MVE组合

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6. 4存在无风险资产的情形

- 定理6.13 存在无风险证券时，如果p是一个MVE组合，那么对于任意组合q我们有

$$\bar{r}_q - r_F = \beta_{q,p}\left(\bar{r}_p - r_F\right)$$

其中 $\beta_{q,p} \equiv \dfrac{cov\left(\tilde{r}_q, \tilde{r}_p\right)}{\sigma_p^2}$

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定理6. 13证明

$$z_p = \frac{\bar{\eta}_p}{\bar{\eta}^T V^{-1} \bar{\eta}} V^{-1} \bar{\eta}$$

$$cov\left(\tilde{r}_p, \tilde{r}_q\right) = z_p^T V z_q = \left(\frac{\bar{\eta}_p}{\bar{\eta}^T V^{-1} \bar{\eta}} V^{-1} \bar{\eta} \right)^T V z_q = \frac{\bar{\eta}_p \bar{\eta}_q}{\bar{\eta}^T V^{-1} \bar{\eta}}$$

取 q=p，有 $\sigma_p^2 = \dfrac{\bar{\eta}_p^2}{\bar{\eta}^T V^{-1} \bar{\eta}}$

$$\beta_{q,p} = \frac{cov\left(\tilde{r}_q, \tilde{r}_p\right)}{\sigma_p^2} = \frac{\dfrac{\bar{\eta}_p \bar{\eta}_q}{\bar{\eta}^T V^{-1} \bar{\eta}}}{\dfrac{\bar{\eta}_p^2}{\bar{\eta}^T V^{-1} \bar{\eta}}} = \frac{\bar{\eta}_q}{\bar{\eta}_p}$$

$$\bar{r}_q - r_F = \beta_{q,T} \left(\bar{r}_T - r_F \right)$$

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